

QPSK Demodulation & Signal Recovery Analysis

Question 1: Are the two data signal's the same? Explain your answer.

No, the two data signals X1 and X2 are not the same. While they are derived from the same source sequence, the Serial-to-Parallel converter separates the bits such that X1 contains the odd-numbered bits and X2 contains the even-numbered bits.

Question 2: Why are the two signals offset in this way? Tip: If you're not sure, see the preliminary discussion for a clue.

They are offset by one bit-period of the master clock because the converter latches the data sequentially. The first bit is held on X1, and the second bit is held on X2. This timing offset ensures that the two parallel streams are synchronized to be processed as a single dibit symbol for the QPSK modulator.

Question 3: What aspect of the signal's shape indicates that the Tunable LPF module's output doesn't correspond with neither the odd nor even bits of the Serial-to-Parallel Converter module's output?

The recovered signals often appear inverted or attenuated compared to the original X1 and X2 streams. If the phase of the local carrier is not perfectly aligned, the LPF output will show a mixture of both signals (crosstalk) or a phase-shifted version of the data, resulting in a waveform that does not cleanly match the original rectangular pulses.

Question 4: What are these two digital signals that are recovered by the Tunable LPF module?

They are the baseband In-phase (I) and Quadrature (Q) components. In an ideal system, these correspond directly to the X1 (odd bits) and X2 (even bits) signals from the transmitter.

Question 5: How does noise affect the appearance of the points on the constellation diagram?

This is achieved through orthogonal detection. By using two local carriers that are 90° apart (sin and cos), the receiver can independently extract either the I or Q data. The LPF then removes the high-frequency products of the multiplication, leaving only the specific baseband stream intended for that branch.

Question 6: Do you think it would matter if the PSK arm of the demodulator recovered the X2 data instead of the X1 data (and hence, the PSK arm recovered the X1 data instead of the X2 data)? There's no need to justify your answer here because you'll test to see whether you're right shortly.

Initial Prediction Yes, it would matter because the Parallel-to-Serial converter at the receiver expects the bits in a specific order to reconstruct the original message correctly. Swapping them would likely result in a scrambled output.

Question 7: Why does it matter that the PSK arm recovers the Serial-to-Parallel Converter modules X1 output and PSK arm recovers the X2 output and not the other way around?

It is a matter of data integrity and sequence. If X2 is processed as X1 and vice-versa, the interleaving process in the Parallel-to-Serial converter will be incorrect. This results in a bit-sequence where every bit pair is swapped, making the recovered message unintelligible even if the individual bits are correct.

Question 8: What's the qualitative error rate of the recovered digital data signal?

The error rate is negligible or zero. The recovered digital signal matches the transmitted sequence perfectly with clean transitions and stable logic levels.

Question 9: What's the qualitative error rate of the recovered digital data signal?

The error rate is low but present. At this moderate noise level, the recovered bitstream generally maintains its shape, but glitches or jitter become visible on the oscilloscope traces. In the constellation diagram, the four points are no longer sharp; they begin to spread out into small, circular clusters. While the system can still distinguish between logic levels most of the time, the margin for error noise immunity is significantly reduced compared to the noise-free state.

Question 10: What's the qualitative error rate of the recovered digital data signal?

The error rate is high. Frequent bit errors are observed as the noise power begins to exceed the signal's amplitude, causing the comparator to misinterpret logic 0 as 1 and vice-versa.

Question 11: What circuit can be used between the Low-pass filters and the Parallel-to-Serial Converters to reduce the error rate due to noise?

A Decision Circuit or Comparator (Level Slicer). By comparing the noisy filtered analog signal against a reference threshold, the comparator squares up the signal, restoring clean digital logic levels and rejecting low-level noise fluctuations before the bits are recombined.